

AZIM PREMJI UNIVERSITY, BHOPAL

First Semester 2026-2027

Tutorial Sheet-1

Course No. MTH-113

Course title: Introduction to Quantitative Thinking

- (1) Compare the following pairs of numbers using $>$ or $<$.
(a) $0 \square -8$; (b) $-1 \square -15$; (c) $5 \square -5$;
(d) $11 \square 15$; (e) $0 \square 6$; (f) $-20 \square 2$

- (2) Write all the integers between the given pairs (write them in the increasing order.)
(a) 0 and -7 (b) -4 and -4 (c) -8 and -15 (d) -11 and 10 (e) -30 and -23

- (3) Suppose your friend says following:
(a) Every positive integer is larger than every negative integer.
(b) Zero is less than every positive integer.
(c) Zero is larger than every negative integer.
(d) Zero is neither a negative integer nor a positive integer.
(e) Farther a number from zero on the right, larger is its value.
(f) Farther a number from zero on the left, smaller is its value.

Do you agree with your friend? Give examples.

- (4) By looking at the number line, answer the following questions:
(a) Which integers lie between -8 and -2 ?
(b) Which is the largest integer and the smallest integer among them?

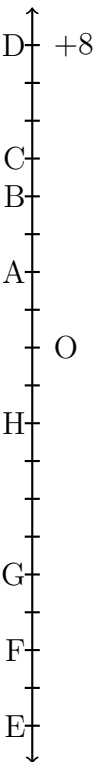
- (5) Write opposite of the following:
(a) Increase in weight; (b) 30 KM north; (c) 326BC;
(d) Loss of Rs 700; (e) 100 m above sea level.

- (6) Draw a number line and answer the following:
(a) Which number will we reach if we move 4 numbers to the right of -2 ?
(b) Which number will we reach if we move 5 numbers to the left of 1.
(c) If we are at -8 on the number line, in which direction should we move to reach -13 ?
(d) If we are at -6 on the number line, in which direction should we move to reach -1 ?

- (7) (a) Write four negative integers greater than -20 .
(b) Write four negative integers less than -10 .

- (8) Adjacent figure is a vertical number line, representing integers. Observe it and locate the following points:

- (a) If point D is $+8$, then which point is -8 ?
(b) Is point G a negative integer or a positive integer?
(c) Write integers for points B and E.
(d) Which point marked on this number line has the least value?
(e) Arrange all the points in decreasing order of value.



- (9) Find the solution of the following:

- (a) $(+7) + (-11)$
(b) $(-13) + (+10)$
(c) $(-7) + (+9)$
(d) $(+10) + (-5)$
(e) $(-217) + (-100)$

- (10) Find the sum

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- (a) $(-7) + (-9) + 4 + 16$
(b) $(37) + (-2) + (-65) + (-8)$

(11) Fill in the blanks with $>$, $<$ or $=$ sign.

- (a) $(-3) + (-6) \square (-3) - (-6)$
(b) $(-21) - (-10) \square (-31) + (-11)$
(c) $(-25) - (-42) \square (-42) - (-25)$

(12) Fill in the blanks

- (a) $(-8) + \square = 0$
(b) $13 + \square = 0$
(c) $\square - 15 = -10$
(d) $(-4) + \square = -12$

(13) Find

- (a) $(-7) - 8 - (-25)$
(b) $(-13) + 32 - 8 - 1$
(c) $(-7) + (-8) + (-90)$
(d) $50 - (-40) - (-2)$